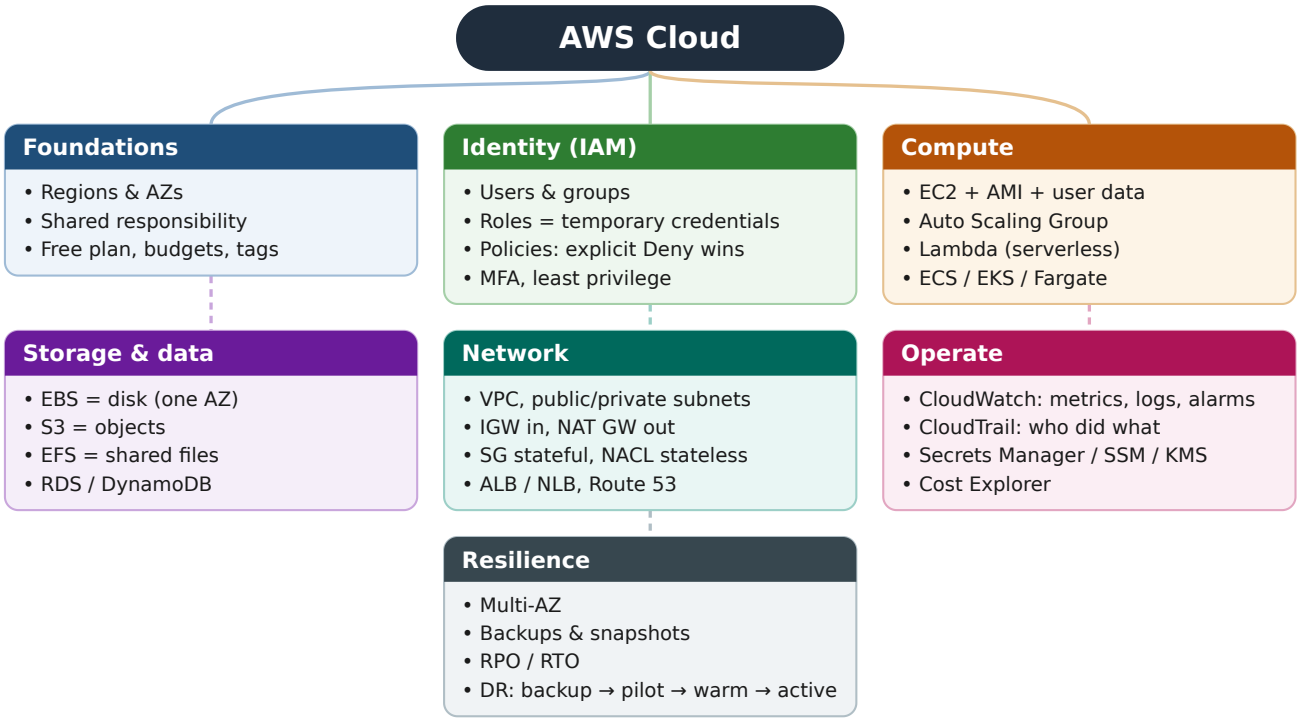
7. AWS Cloud

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7.1 Cloud concepts

Term	Meaning
Cloud computing	Renting IT resources on demand, paying for what you use
IaaS	You get raw servers, networks and disks (EC2, VPC)
PaaS	The platform runs your code; you don't manage servers (Elastic Beanstalk, RDS)
SaaS	Finished software you just use (Gmail, Slack)
Region	A geographic area with its own data centers (`us-east-1` , `ap-south-1` Mumbai)
Availability Zone (AZ)	One or more separate data centers inside a region (`ap-south-1a`). Use 2+ for high availability
Edge location	Small sites close to users, used by CloudFront and Route 53
Shared responsibility	AWS secures the cloud (hardware, buildings); **you** secure what's in it (data, IAM, OS patches, firewall rules)

Term	Meaning
Elasticity / scalability	Grow and shrink resources with demand
High availability (HA)	The system keeps working when one part fails

Pricing models: On-Demand (pay per second/hour), Savings Plans / Reserved (1–3 year commitment, cheaper), Spot (spare capacity, up to ~90% cheaper, can be taken back). **Free plan (new accounts since July 2025):** $100 credit at sign-up, plus up to $100 for completing activities; lasts 6 months or until the credits are used.

7.2 Account safety first

1. Enable **MFA on the root user**, then stop using root.
2. Create an admin identity (IAM Identity Center user, or an IAM user with MFA).
3. Create a **budget** with email alerts (Billing → Budgets).
4. Choose one region and stay in it.
5. **Tag** everything: `Project`, `Owner`, `Env`.
6. After every lab: delete resources and check the Billing console.

7.3 IAM (Identity and Access Management)

Term	Meaning
User	A person or app with long-term credentials
Group	A set of users that share permissions
Role	A set of permissions that is **assumed** temporarily (by EC2, Lambda, CI pipelines, other accounts). No long-term keys
Policy	A JSON document that allows or denies actions
STS	Security Token Service: issues temporary credentials
IAM Identity Center	Central login (SSO) for people, giving short-lived credentials
Least privilege	Give only the permissions that are actually needed

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject", "s3:ListBucket"],
    "Resource": ["arn:aws:s3:::my-bucket", "arn:aws:s3:::my-bucket/*"]
  }]
}
```

Evaluation rule: everything is denied by default → an explicit Allow grants → an explicit **Deny always wins**. **Best practices:** MFA everywhere, roles instead of access keys, no keys in code, rotate or delete unused keys, use IAM Access Analyzer.

7.4 AWS CLI

Command	What it does
`aws configure sso`	Log in through IAM Identity Center (preferred)
`aws configure`	Set up access keys (avoid if possible)
`aws sts get-caller-identity`	"Who am I?" — always check this first
`aws s3 ls`, `aws s3 cp file s3://bucket/`	List buckets / upload
`aws ec2 describe-instances --output table`	List instances
`--profile dev`, `--region ap-south-1`	Choose a profile / region
`--query 'Reservations[].Instances[].InstanceId'`	Filter output (JMESPath)

7.5 EC2 (virtual servers)

Term	Meaning
Instance	A virtual server
AMI	The image the server starts from (Ubuntu, Amazon Linux)
Instance type	Size: `t3.micro` (2 vCPU, 1 GB), `m5.large`. Families: t = burstable, m = general, c = compute, r = memory
Key pair	SSH keys to log in (download the `.pem` once, `chmod 400`)
User data	A script that runs at first boot (install software automatically)
Instance metadata	Info the instance can read about itself at `169.254.169.254` (use IMDSv2)
Instance profile	Attaches an IAM role to EC2, so apps on it get credentials without keys

States: pending → running → stopping → stopped (no compute charge, disks still charged) → terminated (deleted).

```
#!/bin/bash
# user data example
dnf install -y nginx
systemctl enable --now nginx
```

7.6 Storage: EBS, instance store, EFS, S3

Service	Type	Use
EBS	Block disk attached to one EC2 in one AZ	OS disk, databases. Types: gp3 (general), io2 (high IOPS)
Snapshot	Point-in-time backup of an EBS volume, stored in S3	Backups, copying disks to other AZs/regions
Instance store	Temporary disk on the host. **Lost on stop**	Cache, scratch space
EFS	Shared network file system for many instances	Shared files

Service	Type	Use
S3	Object storage, unlimited	Files, backups, static websites, logs

7.7 Security Groups and Elastic IP

- **Security Group (SG):** a stateful firewall around an instance. Allow rules only. Return traffic is allowed automatically. Can reference other SGs (e.g. "allow 5432 from the app SG").
- **Good practice:** SSH (22) only from your IP (or use SSM Session Manager and close 22), HTTP/HTTPS from `0.0.0.0/0` only on the load balancer.
- **Elastic IP:** a static public IPv4 address. All public IPv4 addresses are billed per hour, so release unused ones.

7.8 S3

Term	Meaning
Bucket	A container with a globally unique name
Object / key	A file and its full name (`logs/2026/app.log`)
Storage classes	Standard, Standard-IA, Intelligent-Tiering, Glacier (cheaper, slower to retrieve)
Versioning	Keeps old versions; protects against deletes and overwrites
Lifecycle rules	Move to cheaper classes or delete after N days
Encryption	SSE-S3 (default) or SSE-KMS
Block Public Access	Keep ON unless you truly need a public bucket
Bucket policy	JSON permissions on the bucket
Presigned URL	A temporary link to a private object

7.9 VPC (your private network)

```
VPC 10.0.0.0/16  (one region)
├── AZ-a
│   ├── Public subnet  10.0.1.0/24   → route 0.0.0.0/0 → Internet Gateway   [ALB, NAT GW, bastion]
│   └── Private subnet 10.0.11.0/24  → route 0.0.0.0/0 → NAT Gateway        [app servers]
├── AZ-b
│   ├── Public subnet  10.0.2.0/24
│   └── Private subnet 10.0.12.0/24                                          [RDS standby]
└── Internet Gateway
```

Component	Meaning
VPC	An isolated network in one region, with your chosen CIDR
Subnet	A range inside the VPC, in exactly one AZ
Public subnet	Its route table has a route to an **Internet Gateway**
Private subnet	No direct route from the internet
Route table	Rules for where traffic goes. `local` route = inside the VPC
Internet Gateway (IGW)	Connects the VPC to the internet (both directions)

Component	Meaning
NAT Gateway	Lets private instances reach the internet (outbound only). Sits in a public subnet. **Billed hourly + per GB: delete after labs**
NACL	Stateless firewall on the subnet. Allow and deny rules, numbered order. Return traffic needs rules (ephemeral ports 1024–65535)
VPC endpoint	Private access to AWS services (S3, ECR) without going through the internet/NAT
VPC peering	Connect two VPCs privately (CIDRs must not overlap; not transitive)
Transit Gateway	A hub that connects many VPCs and on-premises networks
Site-to-Site VPN / Direct Connect	Encrypted tunnel / private line to an office or data center
VPC Flow Logs	Records of accepted/rejected traffic: for troubleshooting and security
Reachability Analyzer	Tells you why A cannot reach B

	Security Group	NACL
Level	Instance (network interface)	Subnet
State	Stateful	Stateless
Rules	Allow only	Allow and deny
Evaluation	All rules together	In number order, first match wins

SSM Session Manager: a shell on an instance through AWS, with no SSH port open and no keys. Needs the SSM agent + an IAM role.

7.10 Load balancing and Auto Scaling

Term	Meaning
ALB	Layer 7 (HTTP/HTTPS): path/host routing, TLS termination
NLB	Layer 4 (TCP/UDP): very fast, static IPs
Listener	Port + protocol the load balancer listens on (e.g. 443)
Target group	The servers (or containers) traffic is sent to, with a health check
Health check	Regular request (e.g. `GET /health`); unhealthy targets get no traffic
Launch template	The recipe for new instances: AMI, type, SG, user data, role
Auto Scaling Group (ASG)	Keeps min/desired/max instances, replaces unhealthy ones, scales on metrics
Target tracking policy	"Keep average CPU at 50%"

HA web tier: ALB across 2 AZs → ASG across 2 private subnets → RDS Multi-AZ.

7.11 Route 53 and ACM

Term	Meaning
Hosted zone	The DNS records for a domain

Term	Meaning
Alias record	AWS-specific record pointing to AWS resources (ALB, CloudFront, S3). Free and works at the domain root
Routing policies	Simple, Weighted (A/B), Latency, Failover (with health checks), Geolocation
ACM	Free public TLS certificates for ALB/CloudFront, renewed automatically

7.12 RDS (managed databases)

Term	Meaning
RDS	Managed MySQL, PostgreSQL, MariaDB, SQL Server, Oracle. AWS handles patching and backups
Aurora	AWS's high-performance MySQL/PostgreSQL-compatible engine
Multi-AZ	A standby copy in another AZ with automatic failover (**availability**)
Read replica	A read-only copy for scaling reads (**performance**)
Automated backups / snapshots	Point-in-time restore / manual backups
DB subnet group	The private subnets the database may use

Security: private subnets only, SG allows the DB port only from the app's SG, encryption on, password in Secrets Manager.

7.13 Monitoring and audit

Service	Use
CloudWatch Metrics	CPU, network, disk, custom metrics
CloudWatch Alarms	Trigger on a threshold → SNS email, Auto Scaling, etc.
CloudWatch Logs + Logs Insights	Central logs and log search
CloudWatch Agent	Sends memory/disk metrics and log files from instances (not collected by default)
CloudTrail	Audit log of **who did what** via API/console. Keep it enabled
EventBridge	Reacts to events or schedules (e.g. run a Lambda every night)
SNS	Notifications (email, SMS, other services)
AWS Config	Records resource configuration and checks rules

7.14 Secrets and encryption

Service	Use
Secrets Manager	Stores secrets (DB passwords, API keys), supports automatic rotation. Paid per secret
SSM Parameter Store	Stores config and secrets (SecureString). Standard tier is free

Service	Use
KMS	Creates and manages encryption keys used by S3, EBS, RDS, Secrets Manager

Apps read secrets at runtime using their IAM role. **Never** put secrets in code, AMIs, user data or Git.

7.15 Serverless and containers on AWS

Service	Use
Lambda	Run code without servers, triggered by events; pay per request and duration (max 15 min)
API Gateway	HTTP endpoints in front of Lambda or other backends
ECR	Private container image registry
ECS	AWS container orchestrator: task definitions → services
Fargate	Run containers without managing servers (works with ECS and EKS)
EKS	Managed Kubernetes. Control plane billed hourly + worker nodes

7.16 Cost management

- Budgets + alerts, Cost Explorer, cost allocation **tags**, Free plan credit balance.
- Common surprise bills: NAT Gateways, idle load balancers, unattached EBS volumes and old snapshots, public IPv4 addresses, EKS clusters, large data transfer, forgotten resources in another region.
- Save money: right-size instances, stop dev servers at night, Savings Plans/Spot, S3 lifecycle rules, delete unused resources.

7.17 Backup and disaster recovery

Term	Meaning
RPO	Recovery Point Objective: how much data you can afford to lose (e.g. 15 minutes)
RTO	Recovery Time Objective: how quickly you must be back online (e.g. 1 hour)
AWS Backup	Central backup plans for EBS, RDS, EFS, DynamoDB...

DR strategy	Cost	Recovery speed
Backup & restore	$	Hours
Pilot light (core parts always running)	$$	Tens of minutes
Warm standby (small full copy running)	$$$	Minutes
Multi-site active/active	$$$$	Near zero

Well-Architected Framework pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.

7.18 AWS troubleshooting quick checks

Problem	Check in this order
Can't SSH to EC2	Instance running + status checks → public IP → SG allows 22 from your IP → NACL → route table has IGW → correct key and username (`ubuntu`, `ec2-user`) → key permissions `chmod 400`
Private server has no internet	Route `0.0.0.0/0` → NAT GW → NAT GW is in a public subnet with an IGW route → SG/NACL outbound
ALB shows 502/503	Target group health → app listening on the right port → SG allows ALB → targets → health check path returns 200
App can't reach RDS	Same VPC/peering → DB SG allows the port from the app SG → correct endpoint and port → credentials → DB status
Access Denied	`aws sts get-caller-identity` → which policy applies → explicit Deny? → resource policy (bucket policy) → right region/account

7.19 Multi-account setup (how companies really use AWS)

Service	Purpose
AWS Organizations	Groups many accounts (dev, staging, prod, security, logging) under one management account with consolidated billing
Service Control Policies (SCPs)	Guardrails for whole accounts, e.g. "nobody may disable CloudTrail" or "only these regions". They limit, they never grant
AWS Control Tower	Sets up a well-governed multi-account landing zone automatically
IAM Identity Center	One login for people across all accounts, with permission sets

Why separate accounts: a mistake or breach in dev can't touch prod, clear cost per team/environment, simpler permissions.

7.20 Security services

Service	What it does
GuardDuty	Threat detection from CloudTrail, VPC Flow Logs and DNS logs (e.g. crypto-mining, stolen keys)
Security Hub	Central security findings and best-practice checks
Inspector	Scans EC2, ECR images and Lambda for vulnerabilities
Macie	Finds sensitive data (personal data) in S3
WAF	Web Application Firewall in front of ALB/CloudFront/API Gateway (SQL injection, bad bots, rate limits)
Shield	DDoS protection (Standard is automatic and free)
AWS Config	Records configuration changes and checks compliance rules
IAM Access Analyzer	Finds resources shared outside your account and unused permissions

IAM policy with a condition (example: require MFA):

```
{
  "Effect": "Deny",
  "Action": "*",
  "Resource": "*",
  "Condition": { "BoolIfExists": { "aws:MultiFactorAuthPresent": "false" } }
}
```

7.21 Other services you must recognize

Service	One-line purpose
CloudFront	CDN: caches content close to users, HTTPS, works with S3 and ALB
SQS	Message queue: decouples services (producer → queue → consumer)
SNS	Pub/sub notifications (email, SMS, fan-out to SQS/Lambda)
DynamoDB	Serverless NoSQL key-value database
ElastiCache	Managed Redis/Memcached for caching
EFS	Shared NFS file system
Elastic Beanstalk	Upload code; AWS runs the servers (PaaS)
Systems Manager (SSM)	Session Manager, Run Command, Patch Manager, Parameter Store
CodeBuild / CodePipeline / CodeDeploy	AWS's own CI/CD services
CloudFormation / CDK	AWS's own infrastructure as code (Terraform alternative)
Trusted Advisor	Checks for cost, security and limit issues
Service Quotas	Account limits (e.g. number of vCPUs) — a common cause of failed launches

REMEMBER

- **Public subnet** = route to an Internet Gateway. Private servers reach the internet through a **NAT Gateway**.
- **Security Group**: stateful, allow-only, per instance. **NACL**: stateless, allow + deny, per subnet.
- First command when anything fails: `aws sts get-caller-identity` (who am I, which account?).
- Well-Architected pillars: Operations, Security, Reliability, Performance, Cost, Sustainability.
- After every lab: **delete resources and check Billing**.